```
from airflow import DAG

from airflow.operators.python import PythonOperator

from datetime import datetime


# Task 1 -> Create Employee File


def create_employee_file():


    data = """Rahul,45000

Sneha,52000

Amit,61000

Priya,47000

Kiran,39000"""


    with open("/tmp/employees.txt", "w") as file:

        file.write(data)


    print("Employee file created successfully")
```

```python
# Task 2 -> Read Employee Data

def read_employee_data():

    with open("/tmp/employees.txt", "r") as file:
        records = file.readlines()

    print("Employee Records:")

    for record in records:
        print(record.strip())


# Task 3 -> Calculate Salary Expense

def calculate_salary_expense(ti):

    with open("/tmp/employees.txt", "r") as file:
        records = file.readlines()

    total_salary = 0

    for record in records:

        employee, salary = record.strip().split(",")

        total_salary += int(salary)
```

```python
    print(f"Total Salary Expense = {total_salary}")

    ti.xcom_push(key="total_salary", value=total_salary)
    ti.xcom_push(key="total_employees", value=len(records))


# Bonus Task -> Find Highest Salary

def find_highest_salary(ti):

    with open("/tmp/employees.txt", "r") as file:
        records = file.readlines()

    highest_salary = 0
    employee_name = ""

    for record in records:

        employee, salary = record.strip().split(",")

        salary = int(salary)

        if salary > highest_salary:
            highest_salary = salary
            employee_name = employee

    print(f"Highest Salary = {highest_salary}")
    print(f"Employee = {employee_name}")
```

```python
    ti.xcom_push(key="highest_salary", value=highest_salary)
    ti.xcom_push(key="employee_name", value=employee_name)


# Task 4 -> Generate Salary Report

def generate_salary_report(ti):

    total_salary = ti.xcom_pull(
        task_ids="calculate_salary_expense",
        key="total_salary"
    )

    total_employees = ti.xcom_pull(
        task_ids="calculate_salary_expense",
        key="total_employees"
    )

    highest_salary = ti.xcom_pull(
        task_ids="find_highest_salary",
        key="highest_salary"
    )

    employee_name = ti.xcom_pull(
        task_ids="find_highest_salary",
        key="employee_name"
    )
```

```
    report = f"""
Employee Salary Report

Total Employees = {total_employees}

Total Salary Expense = {total_salary}

Highest Salary = {highest_salary}

Employee = {employee_name}

Status = Processed Successfully
"""

    with open("/tmp/salary_report.txt", "w") as file:
        file.write(report)

    print("Salary report generated successfully")
```

```python
# DAG Definition

with DAG(

    dag_id="employee_salary_processing_dag",

    start_date=datetime(2025, 1, 1),

    schedule=None,

    catchup=False

) as dag:


    # Create Tasks

    task1 = PythonOperator(
        task_id="create_employee_file",
        python_callable=create_employee_file
    )

    task2 = PythonOperator(
        task_id="read_employee_data",
        python_callable=read_employee_data
    )
```

```
task3 = PythonOperator(
    task_id="calculate_salary_expense",
    python_callable=calculate_salary_expense
)

task4 = PythonOperator(
    task_id="find_highest_salary",
    python_callable=find_highest_salary
)

task5 = PythonOperator(
    task_id="generate_salary_report",
    python_callable=generate_salary_report
)

# Task Dependencies

task1 >> task2 >> task3 >> task4 >> task5
```